

Options Greeks Simplified

Delta, theta, gamma, and vega explained in plain English with real examples.

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The Greeks are measurements that quantify how an option's price will change in response to different market conditions — stock price movement, time passing, and changes in volatility. You do not need to calculate these numbers yourself. You need to understand what each one means so you can make better decisions about which options to buy, when to hold them, and when to exit.

DELTA

How Much the Option Moves Per Dollar of Stock Movement

Delta measures how much your option's price changes for every \$1 move in the underlying stock. A call option with a delta of 0.50 gains \$0.50 in value for every \$1 the stock rises, and loses \$0.50 for every \$1 it falls. At-the-money options have a delta of approximately 0.50. Deep in-the-money options approach a delta of 1.0 — they move almost dollar-for-dollar with the stock. Deep out-of-the-money options approach a delta of 0 — they barely move at all on small stock moves.

THETA

Daily Time Decay — The Cost of Holding an Option Overnight

Theta is the amount your option loses in value each calendar day from time passing, all else being equal. If an option has a theta of -0.05, it loses approximately \$5 in value per contract every single day just from the clock ticking — even if the stock does not move at all. Time decay is not linear — it accelerates significantly as expiration approaches. An option with 30 days to expiration loses value slowly. An option with 3 days to expiration loses value rapidly.

GAMMA

The Rate at Which Delta Changes

Gamma measures how quickly delta changes as the stock moves. A high gamma means your delta is changing rapidly — your option is becoming more or less sensitive to stock movements very quickly. Gamma is highest for at-the-money options near expiration. This is why 0DTE (zero days to expiration) options are so explosive in both directions — a small stock move can rapidly change the delta from 0.30 to 0.80, creating massive percentage gains. The same dynamic also creates massive losses when the stock moves the wrong direction.

VEGA

Sensitivity to Changes in Implied Volatility

Vega measures how much your option's price changes when implied volatility (IV) changes by 1%. If an option has a vega of 0.10 and implied volatility increases by 1%, the option gains \$10 per contract in value — even without any movement in the stock price. This is why options buyers want volatility to increase after they buy. The dangerous flip side: if you buy options when implied volatility is extremely high (such as right before a major earnings announcement), and the stock moves in your direction but IV collapses afterward, your option can actually lose value despite being right on direction.

GREEK	WHAT IT MEASURES	HELPS OR HURTS BUYERS	KEY INSIGHT
Delta	Option move per \$1 stock move	Helps — positive delta means profit when stock rises	Higher delta = more participation but more expensive premium
Theta	Daily time decay	Hurts — value lost every day regardless of stock movement	The longer you hold, the more theta costs you
Gamma	Rate delta changes	Helps if moving your way, accelerates losses if not	Near expiration, gamma is highest — moves are amplified both ways
Vega	Sensitivity to IV changes	Helps when IV rises after purchase, hurts when IV falls	Buying high-IV options is expensive — IV crush can destroy value

HOW GREEKS CHANGE BASED ON OPTION TYPE AND TIMING

SCENARIO	DELTA	GAMMA	THETA DAMAGE	VEGA SENSITIVITY
Deep In the Money	0.80 to 1.00	Low	Low	Low
Slightly In the Money	0.55 to 0.80	Moderate	Moderate	Moderate
At the Money	~0.50	Highest	Highest	Highest
Slightly Out of Money	0.20 to 0.45	Moderate	Moderate	Moderate
Deep Out of Money	0.01 to 0.20	Low	Low	Low
30+ days to expiration	Normal range	Moderate	Slow daily decay	High
7 to 14 days to expiration	Normal range	High	Moderate daily decay	Moderate
1 to 3 days to expiration	Extreme sensitivity	Highest	Severe daily decay	Low

TST CORE PRINCIPLE

The most dangerous combination for options buyers is deep out-of-the-money options with less than a week to expiration purchased when implied volatility is elevated. You need a massive stock move in a very short time while paying a premium that simultaneously collapses the moment volatility drops. This describes most ODTE "lottery ticket" option trades. Understand what you are buying before you buy it.

Full options curriculum including real trade examples in the TST Academy Options section at topstocktrading.com